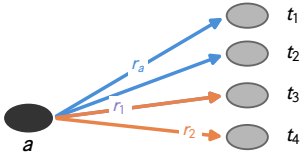
(a) Fixed score-factored operators (existing)

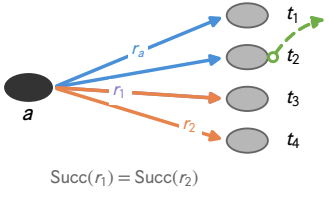


pointwise $\prod$ / $\oplus$ / $(1 - s)$

(no parameters)

identical rankings on $q_1 \neq q_2$ (collision)

(b) Witness-gated algebra (ours, learnable)



witness evidence $\mathcal{W}$

distinct signatures

$\theta_i = g(s_i^{\max}, \bar{w}_i, \rho_i)$

T_Θ gates the family

$\theta \equiv 1 \Rightarrow$ *exact fixed operator (Lemma 1)*

separated rankings $\Rightarrow$ both queries answered